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Soybean Nodule Size and Relationship to Nitrogen Fixation Response to Water Deficit

C. Andy King and Larry C. Purcell*

ABSTRACT

Decreased N_2 fixation in response to drought in soybean [*Glycine max* (L.) Merr.] constrains grain and protein production, but differences exist among cultivars in sensitivity of N_2 fixation to drought. We tested the hypothesis that large nodules may help confer drought tolerance because the fraction of N_2-fixing tissue (i.e., noncortical) is greater in large than small nodules. Consequently, the high energy demand of N_2 fixation may create a greater sink demand by large nodules than small nodules during water deficit, increase phloem water supply to nodules, maintain nodule permeability to O_2, provide sugars to support nodule activity, and supply water for the export of ureides from nodules. This hypothesis was evaluated for the cultivars Jackson (drought tolerant) and KS4895 (drought sensitive) in greenhouse and growth chamber experiments. Individual nodule mass and permeability to O_2 were 0.65 to 0.70 times greater in Jackson than in KS4895 under well-watered and water-deficit conditions. For both cultivars, large nodules maintained a higher relative water content than small nodules across a range of soil-water deficits. One day after labeling leaves of well-watered plants with $^{14}CO_2$, nodules ≤2 mm diam. had approximately 3.5 times the ^{14}C concentration of nodules ≥4 mm diam. For plants of the water-deficit treatment, ^{14}C concentration of nodules ≤2 mm diam. was only 1.6 times that of nodules ≥4 mm diam. Nodules from the plants of the water-deficit treatment had a greater ^{14}C concentration than nodules from the well-watered treatment for all nodule size classes >2 mm diam. Additionally, ^{14}C concentration for all nodule size classes was greater for Jackson than for KS4895. We conclude that drought tolerance of Jackson is partially due to the advantages of large nodules, but that drought tolerance in Jackson also results from an inherently greater supply of photosynthates to nodules.

Soybean cultivars differ in sensitivity of N_2 fixation to water deficits. Incorporating tolerance of N_2 fixation to water deficit in commercial cultivars is a research and production priority (Carter et al., 1999) for two reasons. Firstly, N_2 fixation is decreased in response to water deficit prior to many other physiological processes including leaf gas exchange (Durand et al., 1987; Sall and Sinclair, 1991), C accumulation (Sinclair et al., 1987; Serraj et al., 1997), and soil N accumulation (Purcell and King, 1996). In field studies, seed yield increased in response to fertilizer N in a nonirrigated treatment but there was no response to fertilizer N in irrigated treatments, indicating that improved N_2 fixation during water deficit can increase yields (Purcell and King, 1996). Secondly, high-protein soybean seed production requires a large amount of N, which in most cases is largely derived from N_2 fixation (Matheny and Hunt, 1983).

The cultivar Jackson was identified as the most drought tolerant for N_2 fixation of eight cultivars evaluated (Sall and Sinclair, 1991), and the drought tolerance of Jackson was confirmed in greenhouse (Serraj and Sinclair, 1996; Purcell et al., 1997) and field studies (Serraj et al., 1997). The decrease in N_2 fixation during drought and differences among genotypes in sensitivity of N_2 fixation to water deficits have been associated with one, or a combination of three mechanisms: limited oxygen supply to nodules, nodule C shortage, and potential feedback regulation from accumulation of various N-containing compounds (for review, see Serraj et al., 1999a). A key point linking these three mechanisms is that the nodule water supply is derived almost exclusively from the phloem (Walsh et al., 1989). Decreased volumetric flow in the phloem in response to water deficit may be responsible for eliciting the changes in each of these proposed mechanisms.

Nodule permeability to oxygen (P_o) may regulate nitrogenase activity in response to water deficit (Pankhurst and Sprent, 1975; Weisz et al., 1985) as a result of an increased barrier to O_2 diffusion in the outer cortex of nodules (Pankhurst and Sprent, 1975). Purcell and Sinclair (1995) proposed that, as nodules dried and nodule relative water content (RWC) decreased, an accumulation of solutes in the apoplast of the nodule cortex could cause water to fill intercellular air spaces and decrease nodule P_o. A greater supply of phloem-derived water during water deficits would likely allow greater P_o and continued N_2 fixation by maintaining a high RWC in the nodule cortex (Purcell and Sinclair, 1995) and by providing adequate water for solute export from nodules.

It is not clear whether decreased nodule P_o causes or is a result of decreased nitrogenase activity and nodule respiration in response to water deficits. Nitrogenase activity and nodule respiration decreased prior to a decrease in nodule P_o in response to water deficits (Diaz del Castillo and Layzell, 1995; Purcell and Sinclair, 1995). During the initial stages of water deficit, N_2 fixation activity of the drought-tolerant cultivar, Jackson, and of drought-sensitive 'Biloxi' was completely restored by doubling the ambient O_2 concentration (Serraj and Sinclair, 1996), indicating an O_2 limitation to nodule activity. Interestingly, under well-watered conditions,

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Abbreviations: ARA, acetylene reduction activity; C_o, oxygen conductance; DAS, days after sowing; FTW, fraction transpirable soil water; FWC, fractional water content; NDW, nodule dry weight; NFW, nodule fresh weight; NTW, nodule turgid weight; O_e, external oxygen concentration; P_o, permeability to oxygen; RNR, root plus nodule respiration; RWC, relative water content; SA, surface area; TW, transpirable water.

nodule activity of Jackson was stimulated approximately 50% by elevated O_2 whereas there was no effect of increased O_2 on Biloxi. Although the authors did not measure P_o directly, they speculated that P_o may be less in Jackson than in drought-sensitive cultivars, and through an unknown mechanism contribute to drought tolerance of Jackson. Differences in nodule P_o between drought-tolerant and drought-sensitive cultivars for N_2 fixation have not been directly evaluated.

Differences in photosynthate allocation and water supply to nodules during water deficits, as a result of differences in volumetric flow in the phloem to nodules, may contribute to differential sensitivity of N_2 fixation to water deficits among soybean genotypes. Purcell et al. (1997) proposed that large nodules provided a greater sink for phloem-derived delivery of sugars and water than small nodules. In support of this hypothesis, the relative tolerance of N_2 fixation to water deficit in Jackson was associated with greater individual nodule dry weight than in sensitive cultivars (Purcell et al., 1997, 2000; Serraj and Sinclair, 1998). During moderate drought stress, Jackson had four times the ^{14}C concentration in nodules following exposure of leaves to $^{14}CO_2$ and had twice the acetylene reduction activity (ARA) as the drought-sensitive cultivar KS4895 (Purcell et al., 1997).

The reason that large nodules may have a greater phloem supply than small nodules is that large nodules have proportionately greater amounts of N_2-fixing tissue than do small nodules. Approximately 25% of the volume of a 2-mm-diam. nodule is comprised of infected tissue, whereas a 4-mm-diam. nodule has approximately 60% infected tissue (based on equations found in Weisz and Sinclair, 1988). The high energy demands associated with N_2-fixing tissue may generate a greater sink capacity per unit nodule mass in large nodules, resulting in greater delivery of water and photosynthates to large nodules than to small nodules.

A third potential mechanism associated with decreased N_2 fixation during water deficit is a negative feedback response triggered by elevated levels in the plant of N-containing compounds. It has been proposed that feedback inhibition of N_2 fixation may be in direct response to elevated concentrations of ureides, the products of N_2 fixation, in the nodule (Streeter, 1993). Decreased volumetric flow of phloem supply to nodules would decrease the amount of water available for ureide export in the xylem (Walsh et al., 1989). Nodule ureide concentrations increase in response to water deficits in drought-sensitive cultivars (Purcell and Sinclair, 1995; Serraj et al., 1999b), but no information is available comparing nodule-ureide levels in sensitive and tolerant cultivars in response to water deficit.

Other compounds that may cause feedback inhibition of nitrogenase activity in nodules are amino acids, such as asparagine (Oti-boateng and Silsbury, 1993; Bacanamwo and Harper, 1997; Serraj et al., 1999b;) or glutamine (Neo and Layzell, 1997). Serraj et al. (1999a) proposed that decreased demand for amino acids during water deficit resulted in asparagine accumulation, which has been shown to inhibit leaf ureide breakdown (Lukaszewski et al., 1992). In this case, asparagine transported to nodules would decrease N_2 fixation, and increased leaf ureides would reflect decreased breakdown due to leaf asparagine accumulation. Nevertheless, shoot ureide concentrations have been used as a screening tool to identify soybean genotypes that are potentially drought tolerant for N_2 fixation (Sinclair et al., 2000). Low levels of ureides in shoots of well-watered plants and lower ureide accumulation during soil drying have been documented in Jackson compared to drought sensitive genotypes (Purcell et al., 1998, 2000; Serraj et al., 1997; Serraj and Sinclair 1996; Serraj and Sinclair, 1997), which supports the hypothesis that feedback inhibition may be in response to elevated shoot ureides either directly or indirectly. It appears likely that elevated levels of ureides or other N-containing compounds in soybean roots and/or shoots may play a role in regulating N_2 fixation.

We tested the hypothesis that large nodules in combination with lower leaf ureide levels would confer drought-tolerant N_2 fixation in the cultivar Jackson relative to the drought-sensitive cultivar KS4895. The advantage of large nodules during water deficit was hypothesized to result from a greater phloem supply to nodules, resulting in greater nodule permeability, an increased C and water supply, and a decreased accumulation of ureides in nodules.

MATERIALS AND METHODS

Nodule Gas Exchange Experiment

Jackson and KS4895 were grown in a growth chamber in flow-through pots (Purcell et al., 2000) for nondestructive measurement of ARA and root plus nodule respiration (RNR) in response to soil-water content. The experimental design was a randomized-complete block with four replications of each cultivar by soil-water treatment combination. Pots were constructed of 5-cm-diam. polyvinyl chloride pipe sealed at top and bottom with end caps, and pot volume was 0.5 L. Fittings were inserted into the top and bottom end caps for gas exchange during measurements of ARA and RNR. There was a 20-mm layer of perlite in the bottom of each pot, and pots were filled with a N-free, peat/perlite/vermiculite potting mixture (Basic Mix #2, Sun Gro Horticulture, Garland, TX). Prior to sowing, the potting mix was saturated with deionized water followed by the addition of 500 mL per pot of a N-free nutrient solution (de Silva et al., 1996). Pot-capacity weights were recorded after allowing pots to drain overnight.

Soil was inoculated with *Bradyrhizobium japonicum* (USDA 110) at sowing. A single plant was grown through a hole in the upper end cap of each pot, and plants were kept well watered until initiation of the drying cycle. Each pot received an additional 200 mL of N-free nutrients 7 d before the drying cycle began. The growth chamber was maintained at 24°C with a 16 h photoperiod (0600–2200 h) and PAR of 650 μmol m^{-2} s^{-1} at the top of the plant.

The amount of soil water available for transpiration by plants, transpirable water (TW), at pot capacity was defined as the difference between the pot-capacity weight and the pot weight when daily transpiration for water-deficit plants was $<$10% of the well-watered plants (Ritchie, 1981; Sinclair and Ludlow, 1986). According to this criterion, it was determined in previous experiments that transpirable water was zero at

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22% of pot-capacity weight for both cultivars. The fraction of the total transpirable water (FTW) at a given pot weight was calculated:

$$\text{FTW} = (\text{pot weight} - (\text{pot capacity weight} \times 0.22))(\text{TW})^{-1}. \quad [1]$$

The drying cycle and gas exchange measurements were begun when plants were at the V6 developmental stage (Fehr and Caviness, 1977), approximately 30 d after sowing (DAS). Half the plants from each cultivar were designated either well watered or water deficit, and pots were watered to the desired target weight every 2 h during the photoperiod from 0600 to 2200 h. Well-watered plants were watered to 0.60 FTW, and water-deficit plants were watered to 0.60, 0.24, 0.19, 0.15, and 0.10 FTW on Day 1 through Day 5 of the drying cycle, respectively.

On Days 1 through 4, RNR was measured at 1030 h at 21 kPa O_2 using a closed-system respirometer (Bacanamwo et al., 1997; Bacanamwo and Purcell, 1999) by following the linear depletion of O_2 in the enclosed root system for 10 min. On Day 5, changes in RNR in response to O_2 concentrations surrounding the roots, ranging from 6150 to 10 250 μmol L^{-1}, were used to determine the conductance (C_o, $mm^3\ s^{-1}$) and permeability (P_o, $\mu m\ s^{-1}$) of nodules to O_2 using a one-dimensional model of Fick's first law (Bacanamwo et al., 1997; Bacanamwo and Purcell, 1999). Two assumptions of this model are that O_2 concentration inside the nodule is negligible relative to the concentration surrounding the nodule and that root respiration is constant over the concentration range used for measurements (Bacanamwo et al., 1997). The RNR was regressed against the O_2 concentration surrounding the root system (O_e, μmol $O_2\ mm^{-3}$), and the change in RNR (ΔRNR) with changes in O_e (ΔO_e) was assumed to be due to changes in nodule-linked respiration (Bacanamwo et al., 1997). The slope of this relationship was defined as C_o:

$$C_0 = (\Delta\text{RNR})(\Delta O_e)^{-1}. \quad [2]$$

Nodule conductance, therefore, is a coefficient equating nodule-linked respiration per plant with the O_2 gradient between the air surrounding the nodule and the nodule interior. P_o was calculated as the quotient of C_o and total nodule surface area (SA, mm^2), such that:

$$P_0 = (\Delta\text{RNR})(\Delta O_e \times \text{SA})^{-1}. \quad [3]$$

Nodule permeability is a coefficient equating nodule-linked respiration per unit of nodule surface area with the O_2 concentration gradient between the air surrounding the nodule and the nodule interior. For a given total nodule mass per plant and for similar rates of nodule-linked respiration per plant, a plant with large nodules would have a higher P_o than a plant with small nodules because the total nodule surface area per plant would be less for a plant with large nodules relative to a plant with small nodules.

To evaluate the influence of water deficits on nitrogenase activity, ARA was measured each day using an in situ flow-through assay (Bacanamwo et al., 1997) by introducing a 10 kPa C_2H_2, 21 kPa O_2, and 69 kPa N_2 mixture from a gas mixing device (Parsons et al., 1992) into the sealed root chamber at 200 mL min^{-1}. After 8 min, gas samples were collected from the exhaust of each pot, and C_2H_4 was quantified by gas chromatography. This short-term acetylene exposure prevented the decline in nodule activity (data not shown) associated with long-term acetylene exposure (Minchin et al., 1983). Plants were harvested on Day 5 and separated into shoots, roots, and nodules. Nodules removed from plants on Day 5 were photocopied, and from the photocopy, nodule number and surface area were determined (Weisz and Sinclair, 1988). Plant parts were dried at 65°C for 72 h and dry weights recorded.

Nodule Water Content Experiment

Seeds of Jackson and KS4895 were sown in 15-cm pots, on 1 July 1999 in a greenhouse at Fayetteville, AR (latitude 36° 5′N). The pots contained approximately 1.9 L of N-free potting mix. The potting mix was saturated with deionized water followed by 750 mL of N-free nutrient solution prior to sowing. Pots were allowed to drain overnight and pot-capacity weights were recorded. The potting mix was inoculated with *B. japonicum* (USDA 110) at sowing and plants were thinned to one per pot after emergence. Each pot received an additional 500 mL of nutrients at 20 DAS. Day/night temperatures were 29 and 24 (±3)°C, and natural illumination was supplemented with 1000 W metal halide lamps for a 16-h photoperiod.

Plants were maintained well watered with deionized water until initiation of water-deficit treatments at 23 DAS, when plants were at the V5 developmental stage. During the drying cycle, pots were watered to a specific target weight at 0900 h daily. Well-watered pots were maintained at 0.60 FTW, and water-deficit pots were dried stepwise over a 5-d period. Four plants of each cultivar and water treatment were harvested at 1300 h on each of the final 3 d, corresponding to FTW values for the water-deficit treatments of 0.23, 0.17, and 0.10. Plants were harvested in a laboratory so that nodule fresh weights (NFW) could be recorded immediately. Soil was removed from plant roots, and nodules were plucked prior to separating the root and shoot. Nodules were sorted using 2.4 and 4-mm wire-mesh sieves resulting in three nodule size classes (<2.4, 2.4–4, and >4 mm), and NFW was recorded by size class. Leaves were separated from the stem, and all plant tissues were dried at 65°C for 96 h and dry weights recorded, including nodule dry weights (NDW) by nodule size class.

The RWC of nodules in each size class was measured by first determining the fractional water content of nodules from well–watered plants (FWC_{ww}) (Purcell and Sinclair, 1995):

$$\text{FWC}_{ww} = (\text{NFW} - \text{NDW})(\text{NFW})^{-1}. \quad [4]$$

It was assumed that nodules from well-watered plants were fully turgid, and FWC_{ww} was used to estimate the nodule turgid weight (NTW) for each water treatment and nodule size class:

$$\text{NTW} = (\text{NDW})(1 - \text{FWC}_{ww})^{-1} \quad [5]$$

and from this the RWC of nodules was calculated:

$$\text{RWC} = (\text{NFW} - \text{NDW})(\text{NTW} - \text{NDW})^{-1} \quad [6]$$

By definition, the RWC of nodules from well-watered plants was 1.0.

Ureides were extracted from approximately 25 mg of dried nodule or leaf tissue in 1.25 mL of 0.2 *M* NaOH for 30 min at 100°C, and concentrations were determined colorimetrically using the procedure of Young and Conway (1942) modified as previously described (de Silva et al., 1996). Data were analyzed as a split plot, with cultivar by water regime as the main-plot and nodule size as the subplot.

The experiment was repeated two additional times, and sowing dates for Runs 2 and 3 were 9 Sept. and 28 Oct. 1999, respectively. In Runs 2 and 3, water deficits were established to allow all plants to be harvested on the same day by staggering the initial withholding of water over a 2- or 3-d period. At harvest, FTW values were 0.60, 0.23, 0.17, and 0.10 FTW for Run 2 and 0.60, 0.17, and 0.10 FTW for Run 3. Plant harvests and tissue analysis were the same as for Run 1 of the experiment.

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Carbon 14 Labeling Experiment

Photosynthate transport to soybean root nodules of different size classes was evaluated in a completely random experiment design with a factorial arrangement of two soybean cultivars and two water regimes with six replications. Seeds of Jackson and KS4895 were sown on 2 Feb. 1999 in 3-L pots in a greenhouse at Fayetteville, AR, under conditions similar to those for the nodule water content experiment. Plants were thinned to one per pot after emergence. Each pot received 1 L of N-free nutrient solution at sowing and an additional 0.5 L at 5 wk after sowing.

Pots were well watered until plants reached the V6 developmental stage. At V6, half of the plants for each cultivar were maintained well watered by watering the soil to 0.60 FTW daily, and half of the plants were dried stepwise over a 4-d period to a final soil weight of 0.17 FTW. Pots were weighed and watered daily at 0900 h during the drying cycle and during the ^{14}C labeling and translocation period.

At 1300 h on the day water-deficit treatments reached 0.17 FTW, the uppermost fully expanded trifoliolate leaf of each plant was exposed to $^{14}CO_2$ for 15 min followed by a 24 h translocation period prior to harvest. To administer the $^{14}CO_2$ pulse, a 1-L mylar bag was placed over the source leaf and sealed around the petiole with tape. A scintillation vial was attached to the side of the mylar bag so that $^{14}CO_2$ generated inside the vial would equilibrate inside the mylar bag that was sealed around the treatment leaf. One-half mL of 0.2 *M* NaOH, containing 1.85×10^5 Bq of $NaH^{14}CO_3$ (Sigma Chemical Co., St. Louis, MO), was injected through a septum in the side of the vial, and $^{14}CO_2$ was generated by injecting 4 mL of 1 *M* lactic acid into the vial. After 15 min, 4 mL of 2 *M* KOH was injected into the vial to trap free $^{14}CO_2$, and the bag was removed from the treated leaf after an additional 15 min.

Twenty four h after exposure to $^{14}CO_2$, plants were sectioned into treated leaf, the remaining leaves, stem, root, and nodules. All plant tissue, except nodules, were dried for 48 h at 65°C. Nodules were frozen at −50°C and then freeze dried, so that nodules maintained their harvest size. Freeze-dried nodules were separated into six size classes using wire-mesh sieves. Nodule size classes were <2, >2 to 2.4, >2.4 to 3.3, >3.3 to 4, >4 to 4.7 and >4.7 mm diam.

Dried plant sections were weighed and combusted in a biological material oxidizer (Model OX100, R.J. Harvey Instr. Corp., Hillsdale, NJ). Evolved CO_2 was trapped in 15 mL of a scintillation cocktail that allows for direct counting of ^{14}C (Carbon 14 Cocktail, R.J. Harvey Instr. Corp., Hillsdale, NJ). Radioactivity of all samples was quantified by liquid scintillation spectrometry (Tri-Carb 4530, Packard Instr. Co., Downers Grove, IL).

RESULTS AND DISCUSSION

Nodule Gas Exchange

The ARA and RNR on Day 5 were similar for Jackson and KS4895 under both well-watered and water-deficit conditions (Table 1). Water-deficit stress on Day 5 decreased ARA from 318 to 92 μmol gdw^{-1} h^{-1}, and there was a similar proportional decrease in RNR for plants of the water-deficit treatment. Mean total nodule dry mass ranged from 0.33 to 0.35 g $plant^{-1}$ and was not significantly different between cultivars or water treatments. Jackson, however, had approximately half as many nodules as KS4895, resulting in 65% greater individual nodule mass. Larger individual nodule mass of Jackson resulted in a total nodule surface area per plant that was 41% less than that of KS4895 despite a similar total nodule mass. On Day 1 of the experiment, ARA for the well-watered treatment was similar between cultivars with rates of 72 and 75 μmol h^{-1} $plant^{-1}$ for Jackson and KS4895, respectively. Over the 5-d measurement period, ARA for the well-watered treatment increased to 102 μmol h^{-1} $plant^{-1}$ for both cultivars.

To decrease variability among days of measurement and among plants, acetylene reduction and respiration data were normalized for the initial activity of each plant expressed and as a fraction of the control treatment for each day. This analysis revealed that relative ARA of KS4895 for the water-deficit plants at 0.24 FTW on Day 2 of the drying cycle was less than the value of control plants, but relative ARA for Jackson was not decreased until an FTW of 0.19 on Day 3 (Fig. 1A). Relative ARA was greater for Jackson than for KS4895 at FTW of 0.24 to 0.10 on Days 2 through 5 of the drying cycle. Absolute rates of ARA for the drought treatments were greater for Jackson than KS4895 on Days 2 and 4 (data not shown).

The decline in relative RNR at 21 kPa O_2 paralleled the decline in relative ARA in response to water deficit (Fig. 1B). In contrast to relative ARA, there were no differences between cultivars for relative RNR in response to water deficit. It was hypothesized that slopes of ARA (μmol C_2H_4 $plant^{-1}$ s^{-1}) vs. RNR (μmol O_2 $plant^{-1}$ s^{-1}) would differ for Jackson and KS4895 during water deficit. Despite considerable differences in relative ARA, however, ARA for both Jackson and KS4895 declined similarly as RNR decreased due to water deficit (Fig. 2). The similar efficiency of ARA per unit respira-

Table 1. Nodule and gas exchange characteristics on Day 5 of the nodule gas exchange experiment.

Cultivar	Water treatment	ARA†	RNR†	Nodule						
				Dry weight	Number	Individual mass	P_o†	C_o†	Surface area	O_2 supply
		μmol gdw^{-1} h^{-1}	μmol $plant^{-1}$ s^{-1}	g $plant^{-1}$	No. $plant^{-1}$	mg $nodule^{-1}$	μm s^{-1}	μL s^{-1}	mm^2 $plant^{-1}$	nmol O_2 mm^{-3} s^{-1}
Jackson	**average‡**	**235**	**0.111**	**0.33**	**125**	**2.8***	**7.1***	**16**	**2244**	**15.3***
KS4895	**average**	**175**	**0.106**	**0.35**	**229***	**1.7**	**4.2**	**13**	**3170***	**10.2**
average‡	**well-watered§**	**318***	**0.169***	**0.34**	**203**	**1.9**	**6.5***	**19***	**3082***	**14.9**
average	**water-deficit§**	**92**	**0.047**	**0.34**	**151**	**2.6***	**4.8**	**10**	**2332**	**10.6**

*** Means between cultivars or water treatments were significantly different as determined by an *F* test ($P \leq 0.05$).**
† ARA, acetylene reduction activity; RNR, root plus nodule respiration; P_o, nodule permeability to O_2; C_o, nodule oxygen conductance.
‡ Values were averaged over treatment effects when analysis of variance indicated interaction terms were nonsignificant.
§ Soil in well-watered and water-deficit treatments was at 0.6 and 0.1 fraction transpirable water (FTW), respectively, on Day 5 of the nodule gas exchange experiment.

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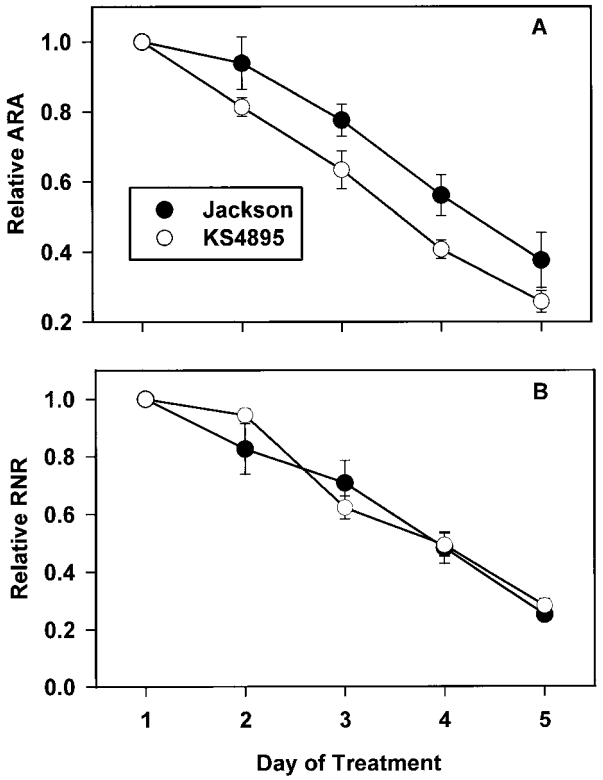


Fig. 1. Relative rates of acetylene reduction activity (ARA) and root plus nodule respiration (RNR) during drought stress. Fraction transpirable-soil water (FTW) for water-deficit treatments on Day 1 through 5 were 0.60, 0.24, 0.19, 0.15, and 0.10, respectively. Error bars represent + SE (n = 4).

tion for Jackson and KS4895 during water deficit may be due to differences in respiration not associated with nitrogenase activity per se. This would include root respiration and nodule maintenance respiration (Bacanamwo et al., 1997). Purcell and Sinclair (1995) found that RNR was decreased by water deficit, and root respiration and nodule respiration were affected similarly. Furthermore, they found that root respiration accounted for 46 to 48% of the total RNR. To resolve if there are differences in the energy costs of N_2 fixation for drought tolerant and sensitive cultivars would require additional experiments separating root from nodule respiration and nodule maintenance respiration from nitrogenase linked respiration. Nevertheless, data in Fig. 2 indicate the strong dependence of nitrogenase activity on respiration during a progressive water deficit for both cultivars.

In that respiration during water deficit was tightly coupled with nitrogenase activity, it was important to establish how the supply of respiratory substrates, O_2 and carbohydrates to nodules, was affected by water deficits in these cultivars. Oxygen supply to support nitrogenase activity on Day 5 was decreased by water deficit as indicated by smaller values of C_o and P_o for plants of the water-deficit treatment than of the control treatment (Table 1). Nodule conductance was similar for Jackson and KS4895 despite the fact that nodule surface area per plant was 30% less for Jackson than for KS4895 (Table 1). To provide O_2 to nodules with

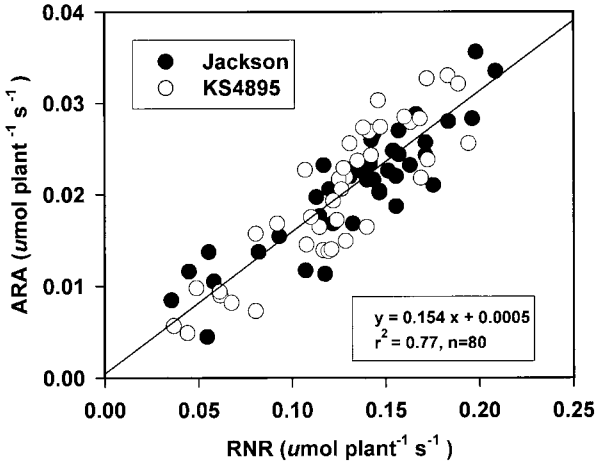


Fig. 2. Acetylene reduction activity (ARA) vs. root plus nodule respiration (RNR). Data are from plants for the water-deficit treatments from the nodule-gas exchange experiment. The regression equation is combined over cultivars.

a much lower nodule surface area per plant, Jackson maintained a greater nodule P_o than did KS4895.

A key question is whether the greater P_o of Jackson was simply a means of compensating O_2 supply per unit nodule surface area or if a greater P_o actually increased O_2 availability to nodules. To address that question, we estimated nodule oxygen supply per unit nodule volume as the quotient of nodule dependent respiration and the sum of individual nodule volumes (mm^3 $plant^{-1}$). Nodule oxygen supply represents the rate of oxygen delivery per unit nodule volume. Nodule dependent respiration at 21 kPa O_2 was calculated by rearrangement of Eq. [2] to solve for a constant respiration rate. The greater P_o of Jackson resulted in the nodule oxygen supply being approximately 50% greater than that of KS4895, averaged over water treatments (Table 1).

Nodule Water Content

Fractional water content of nodules from well-watered plants averaged 0.79 to 0.80 regardless of cultivar or nodule size (data not shown). In Run 1 of the experiment (Table 2), the more drought-sensitive cultivar,

Table 2. Nodule relative water content (RWC) as influenced by fraction transpirable soil water (FTW), nodule size class, and cultivar in the nodule water content experiment.

	Run			Run 1†	
Main effect†	1	2	3	Jackson	KS4895
FTW	relative water content				
0.10	–	0.73b	0.79b	0.70b	0.73b
0.17	–	0.77ab	0.86a	0.68b*	0.75b
0.23	–	0.81a	–	0.80a*	0.89a
Nodule size class					
<2 mm	0.76a	0.70c	0.81b	–	–
2–4 mm	0.73b	0.78b	0.82ab	–	–
>4 mm	0.78a	0.83a	0.84a	–	–

* Indicates significant differences between cultivars within an FTW treatment ($P \leq 0.05$).

† For run 1, there was a significant interaction of cultivar and FTW. Main effects of FTW and nodule size class were significant for runs 1, 2, and 3.

‡ Means followed by the same letter within a run or cultivar are not significantly different as determined by an LSD ($P \leq 0.05$).

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KS4895, had a higher nodule RWC than Jackson at 0.23 and 0.17 FTW, but there was no difference between cultivars at the most severe stress level (0.10 FTW). For Runs 2 and 3 there was no cultivar by FTW interaction for nodule RWC.

Nodule RWC decreased with increasing water-deficit stress to a minimum of 0.70 to 0.79 at an FTW of 0.10, and nodule RWC generally increased as nodule size increased (Table 2). This response of RWC to nodule size was similar between cultivars and among water-deficit treatments, as was indicated by the lack of significance of the interacting effects in the analysis of variance.

Accumulation of ureides in leaves and petioles during water deficits has been hypothesized to result in nitrogenous compounds such as asparagine (Serraj et al., 1999b; Bacanamwo and Harper, 1997) or glutamine (Neo and Layzell, 1997) being sent to nodules and resulting in feedback inhibition of N_2 fixation (de Silva et al., 1996; Serraj et al., 1999b; Purcell et al., 1998, 2000). Furthermore, cultivar differences in drought tolerance of N_2 fixation have been associated with differences in shoot-ureide concentrations, with drought-tolerant cultivars maintaining lower leaf and petiole ureide levels during water deficit (Serraj and Sinclair, 1996; Purcell et al., 1998, 2000; Sinclair et al., 2000). With the exception of Run 2, our results agreed with these previous reports in that Jackson generally maintained lower leaf ureides than KS4895 across water treatments (Table 3). There was a tendency for leaf ureide concentration to increase slightly with increased severity of water deficit (Table 3). Leaf ureides for well-watered plants at 0.60 FTW were significantly less than those at 0.10 FTW in Runs 1 and 3.

Although Jackson tended to have lower leaf ureides than KS4895 (Table 3), nodule ureides were greater in Jackson than in KS4895 at the most severe water deficit (0.10 FTW) in Runs 1 and 3 and for all water-deficit treatments in Run 2 (Table 4). Higher levels of photosynthate delivery to nodules may have allowed the greater nitrogenase activity for Jackson than for KS4895 (Purcell et al., 1997). The continued N_2 fixation of Jackson during water deficit coupled with reduced export of the products of N_2 fixation, ureides, out of the nodules in response to water deficit would account for greater ureide concentrations in nodules of Jackson relative to KS4895 under water-deficit conditions. For well-watered plants (0.60 FTW), there was no difference between cultivars for ureide concentrations in nodules, and concentrations ranged from 8 to 19 μmol gdw^{-1} in the three experiments (Table 4).

Table 3. Response of leaf ureide concentration to main effects of cultivar and fraction transpirable soil water (FTW) in the nodule size experiments.

	Run		
Main effect†	1	2	3
Cultivar	μmol gdw^{-1}		
Jackson	7	27	12
KS4895	10*	30	19*
FTW			
0.10	9ab‡	26a	17a
0.17	8bc	30a	15b
0.23	10a	29a	–
0.60	7c	28a	14b

* Significant differences between cultivars as determined by an *F*-test ($P \leq 0.05$).
† The cultivar by water treatment interaction was not significant.
‡ Means followed by the same letter within a run are not significantly different as determined by an LSD ($P \leq 0.05$).

Table 4. Nodule ureide concentration in response to fraction transpirable soil water (FTW) in the cultivars Jackson and KS4895, averaged over nodule size in Runs 1, 2, and 3 of the nodule size experiment.

		Nodule ureides	
Run†	FTW	Jackson	KS4895
		μmol gdw^{-1}	
1	0.10	34a‡	18b
	0.17	10c	12bc
	0.23	9c	9c
	0.60	9c	8c
2	0.10	35a	25b
	0.17	27b	16c
	0.23	29b	17c
	0.60	15c	12c
3	0.10	42a	32b
	0.17	27b	28b
	0.60	19c	18c

† Means followed by the same letter within a run are not significantly different as determined by an LSD ($P \leq 0.5$).
‡ The nodule size by FTW and nodule size × cultivar interactions were n.s.

Carbon 14 Labeling Experiment

Total radioactivity recovered from the entire plant at 24 h after exposing leaves to $^{14}CO_2$ averaged 1.48×10^6 disintegrations per minute (DPM) and did not differ between cultivars or water treatments (Table 5). The lack of difference in total ^{14}C recovered in plants between water treatments indicates that water-deficit stress was mild with regards to photosynthetic activity. Approximately 60% of the ^{14}C recovered from the plants was present in the treated leaf for all cultivar and soil-water treatment combinations. The major differences in ^{14}C distribution in response to water deficit were in the untreated leaves and nodules. The amount of ^{14}C found in untreated leaves was decreased 70 to 75% in Jackson and KS4895 by water-deficit as compared to well-watered treatment. Both cultivars had an increased percentage of the recovered ^{14}C in nodules of drought-stressed plants, but the increase was greater for Jackson (79%) than for KS4895 (42%). Similarly, the specific activity of ^{14}C in nodules (DPM mg^{-1} nodules) increased more for Jackson (77%) than for KS4895 (35%) in response to water deficit (Table 5). Previous research also indicated that at 24 h after exposing leaves to $^{14}CO_2$, the cultivar Jackson had a greater concentration of ^{14}C in nodules and a greater percentage of the total ^{14}C recovered in the plant in its nodules than did KS4895 (Purcell et al., 1997).

Of particular interest was the potential for preferential allocation of ^{14}C to nodules of different sizes in response to soil-water status. Since the drought-tolerant cultivar Jackson tends to have a greater individual nodule-dry weight and concentrates more photosynthate in nodules during water deficit than KS4895 (Purcell et al., 1997), it was hypothesized that larger nodules may main-

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Table 5. ^{14}C distribution 24 h after pulse labeling with $^{14}CO_2$ in Jackson and KS4895 under control and water deficit (WD) conditions.

Plant section	Jackson		KS4895	
	WD	Control	WD	Control
Total DPM†	1.43 ± 0.08‡	1.42 ± 0.18	1.44 ± 0.16	1.63 ± 0.19
Treated leaf (% Total DPM)	60 ± 2	54 ± 5	64 ± 3	59 ± 3
Untreated leaves (% Total DPM)	1.9 ± 0.1	8.3 ± 1.3	2.1 ± 2.4	6.9 ± 0.4
Stem (% Total DPM)	22 ± 1	24 ± 3	19 ± 1	20 ± 2
Root (% Total DPM)	11 ± 1	11 ± 1	12 ± 1	12 ± 1
Nodules (% Total DPM)	5.2 ± 0.4	2.9 ± 0.4	3.4 ± 0.3	2.4 ± 0.3
Specific nodule (DPM mg^{-1} dw^{-1})	122 ± 13	69 ± 8	81 ± 7	60 ± 7

† Total DPM, total disintegration per minute $\times 10^{-6}$.
‡ SE ($n = 6$).

tain greater metabolic sink strength during drought, contributing to Jackson's superior drought tolerance. In this study, total nodule mass averaged 618 mg $plant^{-1}$ (data not shown) and was not influenced by cultivar or water treatment. Nor was there a cultivar by water treatment interaction or a water treatment main effect on nodule mass within the different size classes. The cumulative fraction of nodule mass, averaged over water treatments, plotted against nodule size class indicates that Jackson had a greater proportion of large nodules than did KS4895 (Fig. 3). The major difference in nodule size between cultivars was in the fraction of nodules > 4.7 mm in diameter, which comprised 18 to 22% of the total nodule mass for Jackson but only 3 to 5% of the nodule mass for KS4895.

The ^{14}C concentration in nodules of various sizes differed between cultivars (Fig. 4A) and water regimes (Fig. 4B), but the cultivar by water regime interaction was not significant. Jackson had 50 to 82% more ^{14}C per unit nodule dry weight than KS4895 for all nodule size classes and soil water regimes. The ^{14}C concentration decreased with increasing nodule size in both cultivars (Fig. 4A) and water regimes (Fig. 4B), with nodules in the two smallest size classes having approximately twice the ^{14}C concentration of those in the three largest size classes. Small nodules may have incorporated a greater percentage of the ^{14}C photosynthate allocated during the 24 h following exposure to $^{14}CO_2$ into structural components than large nodules. In large nodules, more ^{14}C may have been respired or exported as products of nitrogenase activity, thus accounting for the greater ^{14}C concentrations in small nodules.

For nodules < 2 mm in diameter, there was no difference between water-deficit treatments in the ^{14}C concentration in nodules (Fig. 4B). As nodule size increased, however, there was a significantly greater ^{14}C concentration in nodules of the water-deficit treatment than for the well-watered treatment. The hypothesis that small nodules incorporate more C from current photosynthates into structural components and that large nodules lose a greater percentage through respiration and ureide export could explain the difference in ^{14}C concentration between water treatments with increasing nodule size. Under water-deficit conditions, respiration and ureide export are decreased, resulting in increased ^{14}C concen-

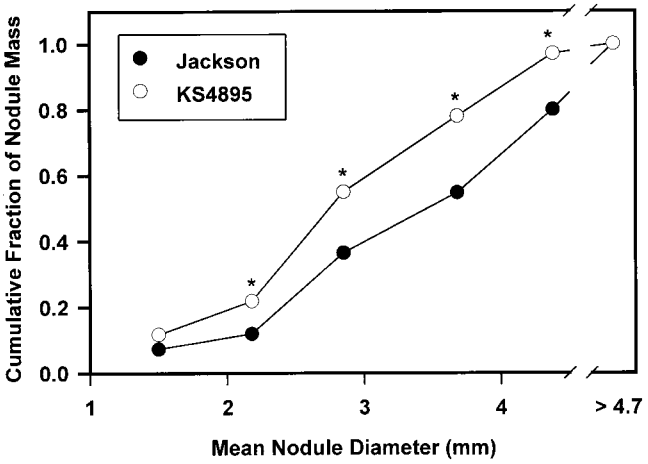


Fig. 3. The cumulative fraction of nodules vs. nodule size for Jackson and KS4895, averaged over water treatments, in the ^{14}C partitioning experiment. * Indicates a significant difference between cultivars at a specific cumulative nodule size as determined by least significant difference ($P < 0.05$).

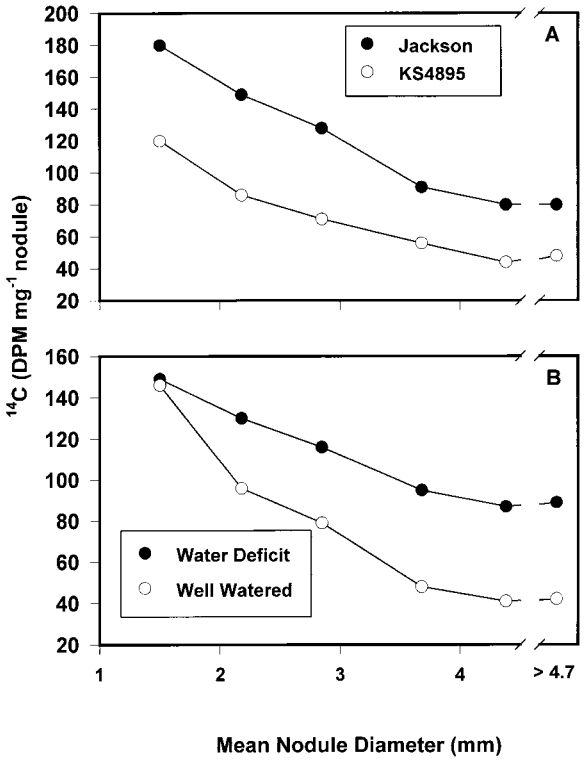


Fig. 4. Specific activity (dpm mg^{-1} $nodules^{-1}$) of ^{14}C in different nodule size fractions as influenced by cultivar, averaged over water treatment (A) and water treatment, averaged over cultivar (B). LSDs ($P < 0.05$) for Fig. 4A within and between cultivars are 14 and 20, respectively, and for Fig. 4B within and between water treatments are 13 and 18, respectively.

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trations in larger nodules as compared to well-watered conditions.

Clearly, the data indicate that large nodule size by itself does not result in preferential photosynthate allocation compared to small nodules. Additionally, increased ^{14}C concentrations in nodules in response to water deficit does not support the hypothesis that N_2 fixation is limited by photosynthate supply during water deficit.

CONCLUSIONS

These experiments confirm the drought tolerance of N_2 fixation of Jackson (Sall and Sinclair, 1991; Serraj and Sinclair, 1996; Purcell et al., 1997) and indicate several fundamental differences between Jackson and the drought-sensitive cultivar, KS4895, that likely confer drought tolerance. First, nodule permeability to O_2 was greater for Jackson than for KS4895, which resulted in a greater O_2 supply to nodules per unit nodule volume. For both cultivars, nitrogenase activity during water deficit was tightly linked to RNR. Although the relationship of acetylene reduction as a function of RNR during water deficit was similar for Jackson and KS4895 (Fig 2), a greater proportion of the total RNR may be used to support N_2 fixation for Jackson than for KS4895. Further work is required to separate root respiration, nodule maintenance respiration, and nitrogenase linked respiration during water deficits in these cultivars.

A second factor contributing to drought tolerant N_2 fixation of Jackson was greater individual nodule size. Larger nodules tended to have greater RWC during water deficits than smaller nodules. Although there was no consistent difference between cultivars in nodule RWC for nodules within a size class, the greater proportion of large nodules for Jackson was a likely advantage in maintaining a favorable nodule water balance.

Thirdly, Jackson had a greater ^{14}C concentration in nodules of all size classes during water-deficit stress than did KS4895 following exposure of leaves to $^{14}CO_2$ (Table 5). The increased ^{14}C concentration in nodules of Jackson relative to KS4895 in response to water deficit (Table 5) may be due to one or a combination of the following factors: (i) decreased nodule respiration, (ii) increased photosynthate allocation to nodules, (iii) decreased export of ^{14}C from nodules in the xylem, and (iv) increased ^{14}C incorporation into nodule structure. Although respiration was certainly decreased by water deficit in both cultivars, it appears unlikely that greater nodule ^{14}C concentration in Jackson was because nodule respiration was decreased more in Jackson than in KS4895. The fact that Jackson maintained higher relative ARA throughout the water deficit in the gas exchange experiment than did KS4895 would indicate that nodule-linked respiration may have been greater for Jackson. In a previous experiment, Jackson had a similar increase in ^{14}C concentration in nodules during water deficit relative to KS4895 as found in this report, and respiration of ^{14}C photosynthates by roots and nodules was greater for Jackson than for KS4895 (Purcell et al., 1997). Associated with the greater amount of respired ^{14}C was a significantly greater ARA activity for Jackson than for KS4895. The greater increase in ^{14}C concentration in nodules of Jackson than of KS4895 during water deficit is presumed due to a greater amount of photosynthate being exported from the leaves and delivered to the nodules.

The role of nodule size on the greater photosynthate allocation to nodules of Jackson than of KS4895 is speculative. Surprisingly, for both cultivars, the ^{14}C concentration was greatest for small nodules (<2.8 mm diam.) under well-watered and water-deficit conditions, but ^{14}C concentrations increased more for large nodules than for small nodules during water deficit (Fig. 4B). The preponderance of large nodules for Jackson appeared to be one factor contributing to a greater ^{14}C concentration during water deficit, but Jackson also appeared to have an inherently greater photosynthate allocation to nodules regardless of nodule size or water treatment (Fig. 4A). Greater O_2 supply and photosynthate allocation to nodules during water deficit likely allowed continued respiratory activity to support N_2 fixation in Jackson compared with KS4895.

Finally, leaf ureides were generally less for Jackson than for KS4895 during water-deficit stress, but nodule ureide concentration was greater. Lower shoot ureide concentrations for Jackson have been noted in several previous reports (Serraj and Sinclair, 1996; Purcell et al., 1998, 2000), and continued ureide degradation in leaves during water deficit has been hypothesized as a key point for preventing feedback inhibition of N_2 fixation in response to delivery of ureides or other nitrogenous compounds to the nodules in the phloem (Purcell et al., 1998, 2000; Serraj et al., 1999a). The fact that Jackson maintained higher nitrogenase activity while nodule ureide concentrations increased during water deficit, relative to KS4895, indicates that feedback inhibition, if it occurs, is more likely triggered by compounds other than ureides that are delivered in the phloem from the shoot, possibly in response to elevated shoot ureide concentrations. Certainly, large nodules of Jackson would favor photosynthate and water allocation, maintain a favorable nodule RWC, and provide continued supply of water for export of ureides in the nodule xylem.

REFERENCES

Bacanamwo, M., and J.E. Harper. 1997. The feedback mechanism of nitrate inhibition of nitrogenase activity in soybean may involve asparagine and/or products of its metabolism. Physiol. Plant. 100: 371–377.

Bacanamwo, M., L.C. Purcell, and R. Serraj. 1997. Closed system O_2 uptake: A simple technique of nodule oxygen permeability estimation. J. Exp. Bot. 48:1987–1994.

Bacanamwo, M., and L.C. Purcell. 1999. Soybean dry matter and N accumulation responses to flooding stress, N sources and hypoxia. J. Exp. Bot. 50:689–696.

Carter, T.E., Jr., P.I. DeSouza, and L.C. Purcell. 1999. Recent advances in breeding for drought and aluminum resistance in soybean. p. 106–125. *In* H.E. Kauffman (ed.) Proc. of World Soybean Res. Conf. VI, Chicago, IL. 4–7 Aug. 1999. Superior Printing, Champaign, IL.

deSilva M., L.C. Purcell, and C.A. King. 1996. Soybean petiole ureide response to water deficits and decreased transpiration. Crop Sci. 36:611–616.

Diaz del Castillo, L., and D.B. Layzell. 1995. Drought stress, perme-

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ability to O_2 diffusion and the respiratory kinetics of soybean root nodules. Plant Physiol. 107:1187–1194.
Durand, J.L., J.E. Sheehy, and F.R. Minchin. 1987. Nitrogenase activity, photosynthesis and nodule water potential in soyabean plants experiencing water deprivation. J. Exp. Bot. 38:311–321.
Fehr, W.R., and C.E. Caviness. 1977. Stages of soybean development. Spec. Rep. 80. Iowa State Univ. Coop. Ext. Ser., Ames, IA.
Lukaszewski, K.M., D.G. Blevins, and D.D Randall. 1992. Asparagine and boric acid cause allantoate accumulation in soybean leaves by inhibiting manganese-dependent allantoate amidohydrolase. Plant Physiol. 99:1670–1676.
Matheny, T.A., and P.G. Hunt. 1983. Effects of irrigation on accumulation of soil and symbiotically fixed N by soybean grown on a Norfolk loamy sand. Agron. J. 75:719–722.
Minchin, F.R., J.F. Witty, J.E. Sheehy, and M. Muller. 1983. A major error in the acetylene reduction assay: decreases in nodular nitrogenase activity under assay conditions. J. Exp. Bot. 37:641–649.
Neo, H.H., and D.B. Layzell. 1997. Phloem glutamine and the regulation of O_2 diffusion in legume nodules. Plant Physiol. 113:259–267.
Oti-boateng, C., and J.H. Silsbury. 1993. The effect of exogenous amino-acid on acetylene reduction activity of *Vicia faba* L. cv. Fiord. Ann. Bot. (London) 71:71–74.
Pankhurst, C.E., and J.I. Sprent. 1975. Effects of water stress on the respiratory and nitrogen-fixing activity of soybean root nodules. J. Exp. Bot. 26:287–304
Parsons, R., J.A. Raven, and J.I. Sprent. 1992. A simple open flow system used to measure acetylene reduction activity of *Sesbania rostrata* stem and root nodules. J. Exp. Bot. 43:595–604.
Purcell, L.C., M. deSilva, C.A. King, and W.H. Kim. 1997. Biomass accumulation and allocation in soybean associated with genotypic differences in tolerance of nitrogen fixation to water deficits. Plant Soil 196:101–113.
Purcell, L.C., and C.A. King. 1996. Drought and nitrogen source effects on nitrogen nutrition, seed growth, and yield in soybean. J. Plant Nutr. 19:969–993.
Purcell, L.C., C.A. King, and R.A. Ball. 2000. Soybean cultivar differences in ureides and the relationship to drought tolerant nitrogen fixation and manganese nutrition. Crop Sci. 40:1062–1070.
Purcell, L.C., R. Serraj, M. deSilva, T.R. Sinclair, and S. Bona. 1998. Ureide concentration of field-grown soybean in response to drought and the relationship to nitrogen fixation. J. Plant Nutr. 21:949–966.
Purcell, L.C., and T.R. Sinclair. 1995. Nodule gas exchange and water potential response to rapid imposition of water deficit. Plant Cell Environ. 18:179–187.
Ritchie, J.T. 1981. Water dynamics in the soil-plant-atmosphere system. Plant Soil 58:81–96.
Sall, K., and T.R. Sinclair. 1991. Soybean genotypic differences in sensitivity of symbiotic nitrogen fixation to soil dehydration. Plant Soil 133:31–37.
Serraj, R., S. Bona, L.C. Purcell, and T.R. Sinclair. 1997. Nitrogen fixation response to water deficits in field-grown 'Jackson' soybean. Field Crops Res. 52:109–116.
Serraj, R., and T.R. Sinclair. 1996. Processes contributing to N_2-fixation insensitivity to drought in the soybean cultivar Jackson. Crop Sci. 36:961–968.
Serraj, R., and T.R. Sinclair. 1997. Variation among soybean cultivars in dinitrogen fixation response to drought. Agron. J. 89:963–969.
Serraj, R., and T.R. Sinclair. 1998. Soybean cultivar variability for nodule formation and growth under drought. Plant Soil 202:159–166.
Serraj, R., T.R. Sinclair, and L.C. Purcell. 1999a. Symbiotic N_2 fixation response to drought. J. Exp. Bot. 50:143–155.
Serraj, R., V. Vadez, R.F. Denison, and T.R. Sinclair. 1999b. Involvement of ureides in nitrogen fixation inhibition in soybean. Plant Physiol. 119:289–296.
Sinclair, T.R., R.C. Muchow, J.M. Bennett, and L.C. Hammond. 1987. Relative sensitivity of nitrogen and biomass accumulation to drought in field-grown soybean. Agron. J. 79:986–991.
Sinclair, T.R., L.C. Purcell, V. Vadez, R. Serraj, C.A. King, and R. Nelson. 2000. Identification of soybean genotypes with N_2 fixation tolerance to water deficits. Crop Sci. 40:1803–1809.
Sinclair, T.R., and M.M. Ludlow. 1986. Influence of soil water supply on the plant water balance of four tropical grain legumes. Aust. J. Plant Physiol. 13:329–341.
Streeter, J.G. 1993. Translocation—A key factor limiting the efficiency of nitrogen fixation in legume nodules. Physiol. Plant. 87:616–623.
Walsh, K.B., N.J. Canny, and D.B. Layzell. 1989. Vascular transport and soybean nodule function. II. A role for phloem supply in product export. Plant Cell Environ. 12:713–723.
Weisz, P.R., R.F. Denison, and T.R. Sinclair. 1985. Response to drought stress of nitrogen fixation (acetylene reduction) rates by field-grown soybeans. Plant Physiol. 78:525–530.
Weisz, P.R., and T.R. Sinclair. 1988. A rapid non-destructive assay to quantify soybean nodule gas permeability. Plant Soil 105:69–78.
Young, E.G., and C.F. Conway. 1942. On the estimation of allantoin by the Rimini-Schryver reaction. J. Biol. Chem. 142:839–853.

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